CSA47 – DEEP LEARNING

ASSIGNMENT – 1

CO1 – MATHEMATICAL FOUNDATIONS AND MACHINE LEARNING PRINCIPLES

Scenario: Healthcare Risk Prediction

A hospital plans to develop an intelligent system to predict disease risk using patient age, blood pressure, glucose level, cholesterol and family history. The answers below analyse the data, select suitable learning approaches and justify decisions using mathematical reasoning.

TASK 1 – Patient Data Representation and Normalization

Example data (three patients):

Patient	Age (yrs)	BP (mmHg)	Glucose (mg/dL)	Cholesterol (mg/dL)
P1	45	130	110	200
P2	60	150	160	240
P3	30	120	90	180

a) Effect of using raw features without centering or normalization

The features lie on very different scales (age 30–60, glucose 90–160, cholesterol 180–240). Without normalization:

- Features with large numeric ranges dominate distance calculations and dot products, so the model under-uses age even if it is clinically important.
- Gradient descent sees an elongated, ill-conditioned loss surface; a single learning rate is unstable for one feature and too slow for another, giving zig-zag and slow convergence.
- Without centering, features have large non-zero means, which couples the bias term with the weights and correlates weight updates.
- Weight magnitudes depend on measurement units rather than importance, so they are hard to interpret.

b) Feature matrix and normalization procedure

The feature matrix X (3 patients × 4 features: Age, BP, Glucose, Cholesterol) is:

$$X = [\ [45, 130, 110, 200], [60, 150, 160, 240], [30, 120, 90, 180]\ ]$$

Procedure: **z-score standardization** (centering + scaling) applied column-wise: $z = (x - \mu)/\sigma$, where $\sigma = \sqrt{[(1/n)\ \Sigma(x - \mu)^2]}$.

Feature	Mean μ	Deviations (x − μ)	Variance	Std σ
Age	45	0, +15, −15	450/3 = 150	12.247
BP	133.33	−3.33, +16.67, −13.33	466.67/3 = 155.56	12.472
Glucose	120	−10, +40, −30	2600/3 = 866.67	29.439
Cholesterol	206.67	−6.67, +33.33, −26.67	1866.67/3 = 622.22	24.944

Normalized matrix Z:

Patient	Age	BP	Glucose	Cholesterol
P1	0.000	−0.267	−0.340	−0.267
P2	1.225	1.336	1.359	1.336
P3	−1.225	−1.069	−1.019	−1.069

Numerical verification: every column sums to 0 (e.g., Age: 0 + 1.225 − 1.225 = 0), so each mean is 0. Variance check for Glucose: $(0.340^2 + 1.359^2 + 1.019^2)/3 = (0.1156 + 1.8469 + 1.0384)/3 \approx 1.000$. Hence all features have mean 0 and variance 1.

Alternative (min–max): $x' = (x - \min)/(\max - \min)$; e.g., Glucose becomes 0.286, 1.0, 0 for P1, P2, P3.

Interpretation: P2 is above the cohort average on every feature ($z > 1.2$), P3 is below average on every feature, and P1 is near the mean. A z-value of 1.3 means 1.3 standard deviations above average for any feature, so features become directly comparable.

c) Importance for distance-based and gradient-based methods

- **Distance-based (k-NN, K-means):** Euclidean distance adds squared differences. A 40 mg/dL glucose difference contributes 1600 while a 15-year age difference contributes only 225, so large-scale features drown out the rest. Normalization gives each feature a fair contribution.
- **Gradient-based (logistic regression, neural networks):** the gradient for weight w_j is proportional to x_j. Unequal scales give gradients of very different size, elongated loss contours and slow or oscillating training. Standardized inputs give a better-conditioned loss surface, faster stable convergence and avoid saturation of sigmoid/tanh units.

TASK 2 – Vector Representation and Matrix Operations

a) Vector formulation

Order the features as (age, glucose, blood pressure, cholesterol). For one patient the normalized feature vector is $x = [x_1, x_2, x_3, x_4]^T$ and the importance weights are $w = [w_1, w_2, w_3, w_4]^T$. The preliminary risk score is the dot product:

$$s = w^T x = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

For n patients stacked in the feature matrix X (n × 4), all scores are obtained at once by the matrix–vector product **s = Xw**, giving an n × 1 vector of scores.

b) Numerical example

Let $x = [0.60, 0.80, 0.70, 0.50]^T$ (age, glucose, BP, cholesterol scaled to 0–1) and $w = [0.20, 0.40, 0.30, 0.10]^T$ (weights sum to 1).

$$s = (0.20)(0.60) + (0.40)(0.80) + (0.30)(0.70) + (0.10)(0.50) = 0.12 + 0.32 + 0.21 + 0.05 = \mathbf{0.70}$$

Interpretation: the score is on a 0–1 scale, so 0.70 indicates fairly high preliminary risk. Glucose contributes most (0.32 of 0.70, about 46%) because it has a high value and the highest weight. High-scoring patients can be flagged for earlier examination. It is a screening score, not a diagnosis.

TASK 3 – Probability and Conditional Probability

Events: D = patient has the disease; D^c = no disease; E = elevated glucose.

Given: N = 1000; P(D) = 200/1000 = 0.20; $P(D^c) = 0.80$; (E and D) = 160; (E and D^c) = 120.

	Elevated glucose (E)	Normal glucose	Total
Disease (D)	160	40	200
No disease (D^c)	120	680	800
Total	280	720	1000

a) P(E)

P(E) = (160 + 120)/1000 = **0.28**. Check: $P(E|D)P(D) + P(E|D^c)P(D^c) = 0.8(0.2) + 0.15(0.8) = 0.16 + 0.12 = 0.28$.

b) P(D | E)

$P(D|E) = P(D \cap E)/P(E) = 160/280 = 0.16/0.28 \approx$ **0.571 (57.1%)**.

c) P(E | D)

$P(E|D) = P(D \cap E)/P(D) = 160/200 =$ **0.80 (80%)**.

d) Healthcare interpretation

P(E|D) = 0.80 says most diseased patients show elevated glucose, while P(D|E) = 0.571 says a patient with elevated glucose has about a 57% chance of disease against a 20% baseline, a risk increase of nearly three times ($0.571/0.20 \approx 2.9$). In contrast, P(D|normal glucose) = 40/720 ≈ 5.6%. Ranking patients by P(D | findings) lets the hospital prioritise those with the highest posterior risk for confirmatory tests, using limited laboratory and specialist resources efficiently. Note that $P(D|E) \neq P(E|D)$; confusing the two is a common error.

TASK 4 – Bayesian Statistics for Disease Classification

Given: P(D) = 0.10, $P(D^c) = 0.90$, P(+|D) = 0.90, $P(+|D^c) = 0.05$.

a) Bayes' theorem

$$P(D|+) = P(+|D)P(D) / [P(+|D)P(D) + P(+|D^c)P(D^c)]$$

- Numerator: $0.90 \times 0.10 = 0.090$
- $P(+) = 0.090 + (0.05 \times 0.90) = 0.090 + 0.045 = 0.135$
- **P(D|+) = 0.090/0.135 = 0.667 (about 66.7%)**

Even with a good test, a positive result means roughly a 2-in-3 chance of disease, not 90%.

b) Roles of prior and likelihood

The prior P(D) is the belief about disease prevalence before the test. The likelihood P(+|D) is how probable a positive result is if disease is present, compared with $P(+|D^c)$ when it is absent.

The posterior is proportional to likelihood × prior. A low prior pulls the posterior down because many positives come from the large healthy group (4.5% false positives versus 9% true positives). A higher likelihood ratio (0.90/0.05 = 18) pushes it up. If the prior were 30%, the posterior would rise to 0.27/(0.27 + 0.035) ≈ 88.5%.

c) Usefulness under uncertainty

Bayesian reasoning combines prior knowledge (prevalence, history, earlier studies) with new evidence in a principled way. It updates beliefs sequentially as new tests arrive (today's posterior becomes tomorrow's prior), gives calibrated probabilities with uncertainty rather than hard labels, copes with small or noisy data through priors, and prevents base-rate neglect, which helps clinicians decide on follow-up tests and treatment.

TASK 5 – Entropy and Information Gain

a) Entropy

$H(S) = -\Sigma\, p_i \log_2 p_i$, with p(high) = 0.4 and p(low) = 0.6.

$$H = -[0.4 \log_2 0.4 + 0.6 \log_2 0.6] = -[0.4(-1.3219) + 0.6(-0.7370)] = 0.5288 + 0.4422 = \mathbf{0.971\ bits}$$

b) Interpretation of uncertainty

Entropy ranges from 0 (perfectly pure, no uncertainty) to 1 bit (a 50/50 split of two classes). A value of 0.971 is close to the maximum, so the dataset is highly mixed and a randomly chosen patient's class is hard to predict.

c) Information gain

$IG(S, A) = H(S) - \Sigma\, (|S_v|/|S|)\, H(S_v)$, the expected reduction in entropy after splitting on feature A. Example for glucose: 50 patients with high glucose (35 high-risk, 15 low-risk) and 50 with normal glucose (5 high-risk, 45 low-risk).

- H(high glucose) = $-(0.7 \log_2 0.7 + 0.3 \log_2 0.3) \approx 0.881$
- H(normal glucose) = $-(0.1 \log_2 0.1 + 0.9 \log_2 0.9) \approx 0.469$
- Weighted entropy = $0.5(0.881) + 0.5(0.469) \approx 0.675$
- **IG = 0.971 − 0.675 ≈ 0.296 bits**

A feature with higher information gain separates high- and low-risk patients better, so it is selected first in decision trees and in feature selection. A feature with IG near 0 has little predictive value.

TASK 6 – Numerical Computation and Distance Measurement

Patient A = (0.70, 0.80, 0.60), Patient B = (0.50, 0.75, 0.90) for (glucose, BP, cholesterol).

a) Euclidean distance

$$d(A, B) = \sqrt{[\Sigma(a_i - b_i)^2]}$$

- Glucose: 0.70 − 0.50 = 0.20, squared = 0.0400
- BP: 0.80 − 0.75 = 0.05, squared = 0.0025
- Cholesterol: 0.60 − 0.90 = −0.30, squared = 0.0900
- Sum = 0.1325, so **d = √0.1325 ≈ 0.364**

b) Distance as similarity

A small distance means similar clinical profiles; a large one means dissimilar patients. A and B differ mainly in cholesterol (68% of the squared distance) and are nearly identical in blood pressure. The maximum possible distance in the unit cube is $\sqrt{3} \approx 1.73$, so 0.364 indicates moderate similarity. This supports k-NN prediction (use outcomes of similar patients) and clustering.

c) Why normalization is necessary

With raw values, glucose (about 90–200 mg/dL) and cholesterol (about 150–300) would dominate the squared differences while BP or age would barely matter. Normalizing to a common range (or z-scores) makes each feature influence distance in proportion to its relative variation rather than its unit, so "similar" is clinically meaningful.

TASK 7 – Bias–Variance Trade-off

a) Identification

- **Model A** (high training and validation error): **high bias**, underfitting.
- **Model B** (very low training error, high validation error): **high variance**, overfitting.
- **Model C** (low, similar errors): good balance, low bias and low variance.

b) Effect of model complexity

Expected error = Bias² + Variance + Irreducible noise. As complexity increases (more parameters, higher polynomial degree, deeper networks), bias decreases because the model can represent more complex relationships, but variance increases because the fitted model changes greatly with different training samples. Total error is U-shaped and the best model lies where the two are balanced. Simple models: high bias, low variance. Complex models: low bias, high variance.

c) Suitable improvements

- **Model A (reduce bias):** increase capacity (more layers/units, higher-degree or interaction features), add informative features such as family history, reduce regularization, train longer, or use a better model family.
- **Model B (reduce variance):** collect more training data, apply regularization (L1/L2, dropout, weight decay), early stopping, simplify or prune the model, use feature selection, ensembling/bagging and cross-validation for tuning.

TASK 8 – Overfitting, Underfitting and Model Capacity

a) Identification

- Simple model that fails to capture disease patterns: **underfitting** (high bias).
- Moderately complex model that performs consistently: **appropriate generalization**.
- Highly complex model, near-perfect on training data but poor on unseen patients: **overfitting** (high variance).

b) Effect of capacity

Capacity is the range of functions a model can represent (e.g., number of parameters or polynomial degree). Low capacity cannot represent non-linear interactions among age,

glucose, BP and cholesterol, so it fails even on training data. Higher capacity allows more complex decision boundaries and relationships to be learned. The right capacity matches the true complexity of the task and the amount of data.

c) Why performance first improves, then overfits

At first, extra capacity removes underfitting: the model learns genuine structure, so training and validation error both fall. Beyond the optimum, the model has more flexibility than the data justifies and begins fitting noise and patient-specific quirks. Training error keeps falling toward zero but validation error rises, so the generalization gap grows and validation error plotted against capacity is U-shaped. Remedies include regularization, more data, early stopping and cross-validation.

TASK 9 – Supervised Learning and Model Evaluation

a) Why supervised

Each historical record has input features (age, BP, glucose, cholesterol, family history) and a known label (diseased or disease-free). The goal is to learn a mapping f: x → y from labelled examples and predict the label of new patients. Since the output is a discrete class, this is **supervised classification**.

b) Training and testing sets

The **training set** is used to fit model parameters (with a validation split or cross-validation to tune hyperparameters). The **testing set** is held out and used only at the end to estimate performance on unseen patients, giving an unbiased estimate of generalization and revealing overfitting. A typical split is 70–80% training and 20–30% testing, stratified by class.

c) Accuracy

Accuracy = correct predictions / total predictions = 92/100 = **0.92 = 92%**.

Interpretation: the model labels 92 of every 100 unseen patients correctly. However, accuracy can mislead on imbalanced data (if only 8% are diseased, always predicting "disease-free" also gives 92%). In healthcare, sensitivity, specificity, precision, F1-score and the confusion matrix should also be reported because a missed case (false negative) is usually costlier than a false alarm.

TASK 10 – Unsupervised Learning and Patient Grouping

a) Why unsupervised learning is suitable

The records have no risk labels, so there is no target to supervise. Unsupervised learning discovers hidden structure, such as groups of patients with similar profiles, directly from the features, and is useful when labelling is costly because it needs specialist review.

b) K-means procedure

1. Normalize the features (age, BP, glucose, cholesterol) and choose the number of clusters K (elbow method or silhouette score).
2. Initialize K centroids (randomly or with k-means++).
3. **Assign** each patient to the nearest centroid using Euclidean distance.
4. **Update** each centroid as the mean of the patients assigned to it.

5. Repeat steps 3–4 until assignments no longer change. The algorithm minimises $J = \Sigma_k \Sigma \|x - \mu_k\|^2$.

Result: groups such as "young, normal values", "middle-aged with raised glucose" and "elderly with high BP and cholesterol".

c) Distance as similarity

Patients are points in feature space; the Euclidean distance $d(x, y) = \sqrt{\Sigma(x_i - y_i)^2}$ measures how alike their profiles are. Small distance means similar and the same cluster; each patient joins the cluster with the closest centroid. Normalization prevents any feature from dominating.

d) Support for clinical investigation

Clusters generate hypotheses: clinicians can examine each cluster's average profile, identify groups with unusually high glucose or cholesterol or high later disease rates, and prioritise screening, lifestyle programmes or follow-up. Cluster membership is not a diagnosis; findings must be validated by clinicians, laboratory tests and labelled outcomes. Clustering supports, but never replaces, medical judgement.

TASK 11 – Comparison of Supervised and Unsupervised Learning

a) Suitable paradigm for each dataset

- **Dataset A** (features + disease labels): **supervised learning** (classification), since correct outputs are available to learn from.
- **Dataset B** (features only): **unsupervised learning** (clustering, dimensionality reduction, anomaly detection), since only the structure of the inputs can be learned.

b) Comparison of objectives and outputs

Aspect	Supervised (Dataset A)	Unsupervised (Dataset B)
Data	Labelled (x, y)	Unlabelled x
Objective	Learn x → y; minimise prediction error	Discover groups, patterns, structure
Output	Predicted class or probability	Cluster assignments, centroids, reduced representation
Evaluation	Accuracy, sensitivity, specificity, F1, AUC	Silhouette score, within-cluster variance, clinical interpretability
Examples	Logistic regression, decision trees, neural networks	K-means, hierarchical clustering, PCA

c) Recommendation

To predict disease status for a new patient, use **supervised classification** trained on Dataset A (e.g., regularised logistic regression or a neural network on normalized features), because the target is an explicit labelled outcome and performance can be measured on a test set. Unsupervised results from Dataset B can complement it, for example by adding cluster membership as an extra feature or by identifying subgroups for further labelling (a semi-supervised pipeline). Supervised learning is the primary choice; unsupervised learning is a supporting tool.

TASK 12 – Design Challenge: Selecting and Adapting a Learning Strategy

Decision: modify and extend (redesign) the strategy. Keep the supervised classifier for risk prediction, and add a scalable preprocessing pipeline and unsupervised clustering for group discovery.

A. Impact Analysis

a) Effect of growing from 10,000 to 1,000,000 records

- **Data representation:** the feature matrix grows from $10^4 \times d$ to $10^6 \times d$ (100× larger) and may not fit in memory, so we need float32/sparse storage, chunked or distributed storage and mini-batch processing.
- **Numerical computation:** pairwise distance matrices ($O(n^2)$, 10^{12} entries) become infeasible; matrix–vector products and gradients cost $O(nd)$ per pass. Floating-point issues (overflow in exp, underflow of probabilities, accumulation error) matter more, so we use log-probabilities, stable softmax/log-sum-exp and standardization.
- **Learning efficiency:** full-batch gradient descent is too slow, so we use mini-batch SGD/Adam and mini-batch K-means with vectorised or GPU computation. More data lowers variance and permits higher-capacity models, but cost per epoch rises roughly linearly.

b) Importance of normalization, missing values and variability

- **Normalization** puts features on a common scale so gradient descent converges and distances are meaningful. At scale, μ and σ are computed on training data (incrementally) and reused for new patients.
- **Missing values:** a million real records will contain gaps. Dropping rows wastes data and may bias results if missingness depends on health status. Use mean/median or model-based (k-NN, regression) imputation plus missing-indicator features, with statistics taken from training data only.
- **Data variability:** differences between wards, devices, age groups and outliers add noise and shift. Robust scaling (median/IQR), outlier clipping and stratified sampling stop extreme values distorting means, distances and gradients.

B. Evaluation of the Original Approach

a) Is a single supervised approach sufficient? No. A supervised model answers "what is this patient's risk?" but does not discover natural patient groups, which is a different objective, and it needs labels that exist for only part of the records. It also does not itself handle scale differences or missing data.

b) Limitations of labelled learning. Labels are expensive and incomplete. Training only on labelled records discards the many unlabelled records that carry information about the feature distribution and possible subgroups. This reduces the effective sample size, increases variance, may bias the model towards patients who were already tested, and misses new patterns or subtypes that were never labelled.

C. Adaptation or Redesign

a) Proposed combined strategy

1. **Preprocess:** clean data, impute missing values and standardize features (statistics from training data) in a batched or distributed pipeline.
2. **Unsupervised stage:** run mini-batch K-means (K by silhouette/elbow) on all 1,000,000 records, labelled and unlabelled, to discover patient groups; optionally reduce dimensions with PCA first.
3. **Supervised stage:** train a classifier (regularised logistic regression or a small neural network) on labelled records, using normalized features plus cluster assignment or distance-to-centroid as extra inputs.
4. **Semi-supervised extension:** pseudo-label high-confidence unlabelled records and retrain; send uncertain or unusual cases to clinicians for labelling.
5. **Deployment:** output a risk probability and a patient-group label for clinician review.

b) Incorporating the mathematical elements

- **Probability:** the classifier outputs P(disease | x), with prevalence as the Bayesian prior; thresholds prioritise patients and cluster-wise disease rates give group-level priors.
- **Distance measures:** Euclidean distance to centroids for clustering and similarity, k-NN checks and outlier detection (far from every centroid).
- **Normalization:** z-score or robust scaling so distances and gradients behave properly.
- **Model capacity:** with 10^6 records a richer model is justified, but capacity is controlled with validation curves, L2 regularization, dropout and early stopping to stay in the good-generalization region of the bias–variance curve.

D. Evidence-Based Defence

a) Experiments and evaluation measures

- Compare the **original approach** (supervised model only, 10,000 records, simple preprocessing) with the **redesigned approach** (preprocessing + clustering features + supervised model, full data) on the same stratified held-out test set, using k-fold cross-validation and repeated runs.
- **Supervised metrics:** accuracy, sensitivity, specificity, precision, F1, ROC-AUC, calibration (Brier score) and log-loss, with emphasis on false negatives.
- **Clustering metrics:** silhouette score, Davies–Bouldin index, within-cluster sum of squares, stability across random seeds, and clinical profiling of outcome rates per cluster.
- **Generalization:** training–validation error gap, learning curves (error vs. training size) and subgroup performance (age, sex, hospital).
- **Efficiency:** training time, memory use, inference latency and scalability.
- **Ablation:** with/without normalization, imputation and cluster features, to show which component adds value.

b) Justification of the final design

- **Mathematical reasoning:** normalization improves conditioning of the loss surface so gradient methods converge faster; distance-based clustering becomes meaningful; Bayesian probabilities give calibrated, interpretable risk.

- **Generalization:** 100× more data lowers variance; regularization and validation-based capacity selection balance bias and variance; unlabelled data improves knowledge of the feature distribution and reduces overfitting to the small labelled set.
- **Computational needs:** mini-batch SGD and mini-batch K-means cost $O(b \cdot d)$ per step rather than $O(n^2)$ or full-batch passes, so the pipeline scales to millions of records.
- **Data characteristics:** the data are mixed in scale, partly unlabelled and partly missing; the combined design handles all of these, whereas the original single supervised approach handles none fully.

Therefore the original strategy should be **modified, not discarded**: keep supervised classification for prediction and add scalable preprocessing and unsupervised clustering to meet the new requirements.